Catalysis Today 442 (2024) 114895



Contents lists available at ScienceDirect

Catalysis Today

journal homepage: www.elsevier.com/locate/cattod





Highly selective hydrodeoxygenation catalyst for sustainable aviation fuel production from used cooking oil

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A R T I C L E I N F O

Keywords:
Sustainable aviation fuel (SAF)
Used cooking oil (UCO)
Hydrodeoxygenation (HDO)
Hydrotreatment
Hydrocracking
Jet-range hydrocarbons
Ni-Mo/Al_2O_3-SiO_2

A B S T R A C T

The production of Sustainable Aviation Fuel (SAF) utilizing used cooking oil (UCO) as a renewable bio-waste resource holds immense potential for promoting sustainability in the aviation sector. In this study, SAF synthesis *via* hydrodeoxygenation of pretreated UCO over Ni-Mo/alumina-silica catalysts followed up hydroisomerization of deoxygenated oil has been investigated in a fixed-bed reactor. The effect of Al_2O_3/SiO_2 ratio on the physico-chemical properties of Al_2O_3-SiO_2 support and on the overall hydrodeoxygenation performance of the Ni-Mo catalyst supported on Al_2O_3-SiO_2 has been investigated. NiMo catalyst supported on $Al2O_3$-SiO_2 in a 70:30 ratio achieved 86.69 % UCO conversion during the hydrodeoxygenation step and exhibited 74.79 % jet range hydrocarbon fuel selectivity under mild conditions (350°C, 1.1 h^{-1} WHSV, 6 MPa, H_2/oil (v/v): 400). The catalyst activity remained stable for >100 hours highlighting its good stability under the experimental conditions investigated. This work demonstrates the potential of tailor-made supported catalysts for highly selective single-step conversion of UCO into jet-range hydrocarbons, offering a significant step towards the production of sustainable aviation fuels.

1. Introduction

As per the International Energy Agency (IEA), the aviation sector accounted for 2 % of global energy-related CO_2 emissions amounting to 800 million tons of CO_2 in 2022. At present, although the aviation sector accounts for a relatively small share of global emissions, it is growing faster than rail, road, or shipping and is one of the most challenging sectors to decarbonise. The environmental impact of petroleum-based jet fuels, primarily due to the significant emission of greenhouse gases (GHG), has prompted a critical examination of alternative, sustainable energy sources for aviation. The petroleum-based aviation turbine fuel (ATF) not only contribute to GHG emissions but also generate pollutants such as NOx, SOx, CO, particulate matter, and volatile organic compounds [1]. On the other hand, aviation fuel production from sustainable sources such as biomass would not only provide the solution by reducing dependency on fossil fuels but also could offer a promising alternative that minimizes GHG emission. The escalating global consumption of aviation fuel, predicted to increase by nearly 38 % from 2008 levels to 2025 [2], underscores the urgency to explore sustainable alternatives. Biomass-derived aviation fuel has the potential to substitute approximately 30 % of total aviation fuel, presenting a long-term solution to address the aviation industry challenges [3]. In addition, bio-jet fuel has the potential to reduce GHG emissions by 65–85 % compared to petroleum-based jet fuel, which will support mitigating environmental impact from the aviation sector [4]. Various pathways, including the catalytic Sugar-To-Jet (STJ), gasification followed by Fischer-Tropsch Synthesis (FTS), Alcohol-To-Jet (ATJ), and Hydroprocessed Renewable Jet (HRJ) have been explored for manufacturing sustainable aviation fuel [5–8]. Among these routes, HRJ stands out as a suitable method for converting UCO and/or non-edible oil or animal fat-based feedstocks into aviation fuel. This process involves hydrodeoxygenation and hydrocracking of triglycerides in UCO, transforming them into straight-chain alkanes followed by catalytic hydrocracking and hydroisomerization in the subsequent step to meet the required standards for jet fuel. Achieving desirable cold flow properties while retaining the fuel's flash point necessitates the composition of straight-chain alkanes and isomer alkanes within the carbon number range of 9–15.

The current methodologies utilized in the manufacturing of bio-jet fuel encounter a plethora of hurdles, primarily stemming from the

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https://doi.org/10.1016/j.cattod.2024.114895
Received 5 February 2024; Received in revised form 6 June 2024; Accepted 17 June 2024
Available online 23 June 2024
0920-5861/© 2024 Published by Elsevier B.V.

intricate selection process of feedstock and the demanding pretreatment conditions necessary due to the diverse nature and properties of vegetable oils [9,10]. However, amidst these challenges, UCO emerges as a promising solution. Enriched with free fatty acids, UCO stands as a cost-effective and environmentally sound alternative. Widely available across the globe, UCO presents itself as a readily accessible resource, often more economically feasible than pure vegetable oils. Leveraging UCO not only holds the potential to substantially decrease manufacturing expenses but also offers a sustainable solution to address the environmental repercussions associated with the disposal of spent oils into drainage systems [11]. The hydroconversion including hydrodeoxygenation of the UCO followed by hydrocracking and hydroisomerization of the generated paraffins involves a complex process of saturation, heteroatom removal, cracking, and isomerization, requiring bi-functional catalysts. The sulfided NiMo, known for its high hydrogenation and hydrodesulfurization activity, is commonly employed in the hydrotreatment of refinery sulfur-containing feed to produce low and/or nil sulfur products such as naphtha, middle distillate viz kerosene, diesel, etc [12–15]. Furthermore, the choice of support material is very important for any process that involves supported catalysts. A carrier is usually a high-surface area inorganic material containing a complex pore structure through which catalytic materials are deposited. Earlier it was thought that support only provides a surface to disperse the catalytic substance to maximize the catalytic surface area. However, it is now quite clear that support can and often does play a critical role in maintaining the activity, selectivity, and durability of the finished catalyst. By far the most common carriers used in heterogeneous catalysts are non-reducible inorganic oxides with a high surface area such as Al_2O_3, SiO_2, TiO_2, Al_2O_3-SiO_2, ZrO_2 etc. Al_2O_3 and SiO_2owing to their excellent thermal and hydrothermal stability, low cost, and availability in desired purity with a well-controlled range of acidity are frequently employed as catalyst support. To facilitate easy access of bulk molecules to the active sites located within the pores, the support must exhibit a high surface area and large pore size. By changing the Al_2O_3/SiO_2 ratios, Al_2O_3-SiO_2 supports can be designed to meet the above objectives.

Despite the advancements in the hydroprocessing of UCO, there is a gap in understanding the activity and selectivity of sulfided catalysts towards the production of jet-range (C_9-C_{15}) hydrocarbons through complex hydroconversion process chemistry. The previous studies have explored the catalytic hydroprocessing of UCO, revealing insights into conversion rates and product yields [16–24]. However, efficient catalysts with extended lifetime and stability, especially for single-step HRJ production from UCO, remain underexplored. In addressing these gaps, the present study focuses on the hydroconversion of UCO over Ni-Mo/Al_2O_3-SiO_2 catalysts in a fixed-bed reactor. Phosphorus is used as the second promoter in UCO hydroconversion catalyst. Phosphorus, a non-metallic element often overlooked in catalyst studies compared to metallic additives, plays a significant role in the NiMo/Al_2O_3-SiO_2 catalyst system. Research has shown that phosphorus can stabilize the structure of Al_2O_3 and prevent the phase transition of γ-Al_2O_3 to α-Al_2O_3 at high temperatures [25]. When incorporated, phosphorus decorates the hydroxyl groups on the alumina surface, introducing P-OH groups [26] that alter the distribution of Lewis acid and Brønsted acid sites on the Al_2O_3 surface. This modification can have a profound impact on the catalytic properties and performance of the NiMo/Al_2O_3-SiO_2 catalyst, influencing its activity and selectivity in the hydrodeoxygenation process for converting feedstocks like used cooking oil into sustainable aviation fuel range hydrocarbons. The presence of phosphorus leads to an increase in the catalytic activity and reduction of deactivation [27]. The primary objective of the present study is to investigate the catalyst's suitability in producing jet-range hydrocarbon molecules *viz*. paraffins, cycloparaffins, low aromatics with high yield, aligning with jet fuel specifications. This study not only contributes valuable information for optimizing hydrodeoxygenation catalyst formulation and hydroconversion process conditions but also introduces suitable hydroprocessing routes for producing Jet A-1 range aviation fuel from UCO ensuring longer lifetime and stability of catalyst.

2. Experimental methods

2.1. Materials

Ortho-Phosphoric acid 85 % (AR) (85 % H_3PO_4), Spectrochem Pvt Ltd, Ammonium hepta molybdate 99 % (Aldrich Chemicals) ($(NH_4)_6Mo_7O_{24}.4\ H_2O$), Nickel nitrate hexahydrate (AR grade), Aluminum chloride hexahydrate (AR) ($AlCl_3{\cdot}6\ H_2O$), tetraethoxysilane (TEOS) (AR), ethanol (AR) (C_2H_5OH) and Hydrochloric acid (AR) (HCl) were purchased from S.D fine chemicals. Ltd and used as received for the synthesis. For the ZSM-22 synthesis 1,6-diaminohexane $NH_2(CH_2)_6NH_2$, potassium hydroxide (KOH), aluminum sulfate ($Al_2(SO_4)_3{\cdot}18\ H_2O$), and colloidal silica solution (LUDOX AS-40),were obtained from Alfa Aesar, Merck (India) and Sigma-Aldrich respectively. Distilled water was used for making metal solutions and for catalyst preparation. Iso-octane (extra pure, purity > 99 %) procured from Finar Chemicals. Sulfiding agent dimethyldisulphide (>99 % pure) procured from Central Drugs House Pvt. Ltd. Used cooking oil (UCO) used for experimental purpose was palm oil derived UCO procured from M/s. Advaith Biofuels Ltd. (India).

2.2. Catalysts Preparation for Hydrodeoxygenation of used cooking oil

The preparation of the catalyst for hydrodeoxygenation involves a two-stage process. The first stage focuses on synthesizing a porous support material. This support is composed of alumina (Al_2O_3) and silica (SiO_2) with tunable Al/Si ratios. The sol-gel method is employed for this purpose, involving three distinct steps. and the second stage involves depositing nickel (Ni) and molybdenum (Mo) onto the prepared support to get the final catalyst.

In the first stage, Al_2O_3-SiO_2 support synthesis process involved three steps: (1) the synthesis of mixed sol; (2) the synthesis of wet gel and the process of aging treatment; (3) Supercritical ethanol drying. In step 1, alumina sol and silica sol were prepared separately by stirring the mixture of $AlCl_3{\cdot}6\ H_2O$/EtOH/ H_2O = with the mole ratio of 1/12/35 and TEOS/H_2O/EtOH/HCl = with the mole ratio of $1/4/6/7.5\times10^{-3}$ respectively. Then both the sols were mixed in molar ratios of Al/Si = 1/0, 2/1, 3/1 at 30°C for 60 min to complete the hydrolysis. In step 2, 1,4-dioxane was added to the mixed sol to promote condensation. This was done by quickly stirring the mixture of sols at 30°C for 15 min and sealed at room temperature to form the wet gel. In the process of aging treatment, the wet gel was soaked in anhydrous ethanol at 50°C to strengthen the network and swap out the redundant water and organic matter.

In step 3, the wet gel was placed in an autoclave which was sealed and filled with anhydrous ethanol. To ensure safety requirements, the autoclave must be recharged with pure nitrogen gas to exhaust the oxygen. The absolute pressure was controlled in the range of 0.1–9.1 MPa with the temperature increased to 270 °C at a rate of 2 °C/min. After the heat-preservation stage, the absolute pressure was slowly reduced to 0.1 MPa by the slow release of the ethanol. Finally, the samples were obtained at room temperature. The serial numbers of samples, 90Al-10Si, and 70Al-30Si, corresponded to different mole ratios of Al/Si. Al_2O_3–SiO_2 aerogel, samples were calcined at 550 °C respectively for 5 h with a rate of 3 °C/min.

In the second stage for the preparation of the hydrodeoxygenation catalyst, the support γ-alumina (Trilobe extrudates) was first dried at 150°C for 2 hours and calcined at 550°C for 5 hours to remove moisture. The pore volume was measured using the water-wetting technique. The Trilobe extrudates were then impregnated with a 0.5 wt% phosphorus solution (85 % H_3PO_4) using the incipient wetness method. This solution was sprayed onto the extrudates and air-dried for 12 hours at room temperature, followed by drying at 120°C for 12 hours in an oven, and then calcined at 550°C for 5 hours to produce a phosphorus-impregnated trilobe catalyst. Subsequently, 10 wt% molybdenum was

impregnated onto the phosphorus-impregnated calcined trilobe support using an $(NH_4)_6Mo_7O_{24}{\cdot}4\ H_2O$ solution. The sample was air-dried and oven-dried as previously described, followed by calcination at 550°C for 5 hours to obtain MoO_3 oxide-supported P-trilobe.

Next, a 3 wt% nickel nitrate aqueous solution ($Ni(NO_3)_2{\cdot}6\ H_2O$) was impregnated onto this material. The material was dried at room temperature for 12 hours, then at 110°C for 12 hours in an oven, and finally calcined at 550°C for 5 hours in a muffle furnace, resulting in the NiMo/γ-Al_2O_3 catalyst. The same procedure was applied to prepare NiMo/Al_2O_3-SiO_2 catalysts with the 90:10 and 70:30 Al_2O_3-SiO_2 supports.

2.3. Preparation of hydroisomerization catalysts

The preparation of hydroisomerization catalysts involves two main stages, the synthesis of ZSM-22 zeolite framework and the subsequent incorporation of platinum on template-free ZSM-22 zeolite.

ZSM-22 samples (bulk molar SiAl = 45) were synthesized using 1,6-diaminohexane, $NH_2(CH_2)_6NH_2$ (Alfa Acer, 99 %), as a structure directing agent using the method reported elsewhere [28]. As per the method, the crystallization of ZSM-22 was performed using gel molar composition of $27NH_2(CH_2)_6NH_2/12\ K_2O/Al_2O_3/60SiO_2/3670\ H_2O$ by employing potassium hydroxide, KOH (Merck, India); aluminum sulfate, $Al_2(SO_4)_3{\cdot}18\ H_2O$ (Merck, India); and colloidal silica solution (LUDOX AS-40, Sigma-Aldrich) as precursors. Hydrothermal crystallization of the prepared gel was performed at 160°C under stirred conditions for 18 hours. The crystallized samples were then filtered, washed multiple times with deionized water, dried at 110°C for 24 hours, and finally calcined at 550°C for 12 hours in the presence of air to obtain the template-free ZSM-22 zeolite. The template-free zeolite was first ion-exchanged to its ammonium form using a 1 M ammonium nitrate solution under reflux conditions for 24 hours. The H-form of the zeolite was then obtained by calcining the NH_4^+-zeolite at 600°C for 6 hours under an airflow. The zeolite samples were subsequently loaded with 0.5 wt% platinum using tetramine platinum (II) nitrate through an ion-exchange route, maintaining the solution pH around 10. The Pt-loaded zeolite sample (65 wt%) was then mixed with 35 wt% pseudo-boehmite and extruded into cylindrical extrudates of 1.6 mm diameter and approximately 3 mm length. These extrudates were calcined at 400°C under a continuous flow of pure oxygen for 5 hours. The resulting Pt/H-ZSM-22 hydroisomerization catalyst had a bulk molar Si/Al ratio of 45.

2.4. Catalyst characterization techniques

The supports and catalysts were characterized by different techniques. The nitrogen (N_2) adsorption-desorption profiles of catalysts were measured using a Micromeritics ASAP-2020 plus surface area and porosity analyzer to derive the Brunauer–Emmett–Teller (BET) surface area (SBET), while the total pore volume (Vp) and average pore diameter were evaluated using the Barrett–Joyner–Halenda method. The X-ray diffraction (XRD) patterns were recorded on a Panalytical AERIS X-ray diffractometer with Cu Kα emission (λ = 1.5406 Å) operating at 40 kV and 40 mA. The framework acidity of all the samples was estimated using ammonia temperature-programmed desorption (TPD) technique (Micromeritics Autochem Chemisorption analyzer). Typically, the samples were activated under helium flow at 300 °C for 1 h, followed by adsorption of ammonia at 100 °C using a gas mixture of 6 % ammonia in helium at a constant flow rate of 25 mL/min. A higher adsorption temperature was selected to minimize the extent of physical adsorption. Furthermore, to ensure the removal of any physically adsorbed ammonia, the sample was purged at 150 °C for 2 h under a flow rate of 50 mL/min of high-purity helium (99.99 %). TPD of ammonia was performed by heating the sample in the temperature range 150–900 °C, at a rate of 10 °C/min and under a helium flow rate of 25 mL/min. The amount of ammonia desorbed from the sample was determined using a thermal conductivity detector (TCD) and was used to quantify acidity in terms of micromoles of ammonia desorbed per gram of zeolite sample.

2.5. Pretreatment of the used cooking oil (UCOs)

Used cooking oil (UCO) underwent a multi-step purification process to remove impurities and Free Fatty Acids (FFAs). The initial step involved filtering the untreated Used Cooking Oil (UCO) to eliminate waste residue, followed by centrifugation to separate the oil layer using a washing technique (hot water extraction method). Subsequently, the oil was aerated to induce bubbling in a separation funnel, facilitating the separation of FFAs from the oil layer. The resulting water/oil mixture underwent phase separation, yielding three distinct phases (water, FFAs, and oils).

Finally, the purified oil layer, devoid of FFAs, was isolated and subjected to evaporation to obtain the pretreated UCO, referred to simply as UCO for the subsequent hydrodeoxygenation reactions. This meticulous pretreatment process ensures the removal of impurities and prepares the UCO for efficient utilization in further chemical transformations.

2.6. Evaluation of catalytic activity

2.6.1. Hydrodeoxygenation of UCO

2.6.1.1. Catalyst loading and evaluation methodology. Hydrodeoxygenation of Used Cooking Oil (UCO): Hydrodeoxygenation of UCO was performed using a fixed-bed, up-flow reactor of dimensions 47 cm × 1.1 cm (internal diameter) with a catalyst bed length of 25 cm and equipped with a thermowell (diameter = 3 mm, length = 420 mm) for measurement of the catalyst bed temperature. The catalyst was placed in the reactor to have a minimum temperature gradient and to have a sufficient layer of inert material towards the input to ensure proper preheating and distribution of feed before it first reached the catalyst zone. During the loading process, 80−120 mesh sand was packed into the void spaces of the catalyst bed, thus ensuring the uniformity of gas/liquid flow distribution. Typically, about 20 g of catalyst extrudates were loaded in the reactor as per the procedure.

2.6.1.2. Catalyst drying. After loading the catalyst in hydrodeoxygenation (HDO) reactor, the temperature of the catalyst bed is raised from room temperature to 120 ^{0}C at 1^0 C/ min under the gentle flow (0.6 SLPH/gm of catalyst) of hydrogen. Drying of the catalyst is carried out at 120 ^{0}C and 0.1 MPa hydrogen pressure while maintaining the gentle flow of hydrogen for 4 hr to remove any moisture in the catalyst or the reactor system.

2.6.1.3. Catalyst wetting. The wetting step looks contradictory after carrying out drying in the first step, however, the wetting of the catalyst is carried out with neat hydrocarbons such as iso-octane. The pressure of the reactor system is increased gradually to 4 MPa at the prevailing catalyst bed temperature 120 ^{0}C under gentle hydrogen flow (0.6 SLPH/gm of catalyst). An exotherm is seen during the wetting step resulting from the heat of absorption that is released as hydrocarbon enters the pores of the extremely dry catalyst. Wetting ensures that all the catalyst particles are evenly wet with hydrocarbon, leading to improved distribution once the sulfiding step begins. A sufficient flow rate of iso-octane (~ 5 mL/h/g cat) is desirable during wetting to get better feed distribution and for fast removal of the heat that is formed during absorption. However, depending on reactor length and internal diameter, neat iso-octane flow can be adjusted to ensure complete catalyst wetting in the hydro-deoxygenation reactor.

2.6.1.4. Catalyst sulfidation. Once wetting of all the catalyst particles is ensured, the reactor system is made ready to start sulfidation.

Sulfidation is carried out at 4 MPa under a hydrogen flow of (0.6 SLPH/g of catalyst) under varying temperature conditions. The step-by-step procedure was adopted to complete sulfidation of Hydro-deoxygenation catalyst.

1. The catalyst bed temperature was increased from drying temperature of 120 ^{0}C to 240 ^{0}C at ramp rate of 1^0 C/ min while maintaining the neat iso-octane flow (~ 5 mL/h/g cat). The neat iso-octane feed was stopped once the catalyst bed temperature reached 180 °C and injection of iso-octane spiked with DMDS (2.5 mL/h/g cat) was started. The concentration of DMDS was maintained at 0.6 g/L of iso-octane for each gram of catalyst. The sulphidation was continued till the temperature reached 240 °C and maintained at 240 °C till the time H_2S concentration reached 0.5 % or more at the outlet. If H_2S is not present then sulfidation is carried out for 4 h at 240 °C.
2. Then the catalyst bed temperature was increased to 350 °C at 1 °C /min with continued DMDS feed injection. The temperature was maintained at 350 °C till the time H_2S concentration reached 1 % or more at the outlet. If H_2S is not present then sulfidation is carried out for 4 h at 350 °C.
3. The reactor pressure is gradually increased to 6 MPa by maintaining the catalyst bed temperature at 350 °C. Once 6 MPa hydrogen pressure is achieved at 350 °C bed temperature, the UCO feed to the reactor with H_2/Oil ratio (v/v) 400 is started.

2.6.1.5. Product analysis. The product from the reactor outlet was cooled using a heat exchanger connected to a chiller, and it was further separated into gas and liquid phases in a separator. The liquid product was collected and allowed to settle in a separating funnel from where water formed in the reaction was separated. The hydrocarbon product was dried overnight over anhydrous sodium sulfate and analysed for various physical properties. The viscosity and pour points were measured as per ASTM D445 & D97 respectively. SIMDIST analysis was performed on liquid products to ascertain the different ranges of hydrocarbons formed during the reaction.

2.6.2. Hydroisomerization of the hydrodeoxygenated oil

This was performed using a fixed-bed, up-flow reactor of dimensions 47 cm × 1.1 cm (internal diameter) with a catalyst bed length of 8 cm and equipped with a thermowell (diameter = 3 mm, length = 40 mm) for measurement of the catalyst bed temperature. The catalyst was placed in the reactor to have a minimum temperature gradient and to have a sufficient layer of inert material toward the input to ensure proper preheating and distribution of feed before it first reached the catalyst zone. During the loading process, the catalyst was typically divided into multiple batches. Following each batch of the catalyst, 80−120 mesh sand was packed into the void spaces of the catalyst bed, thus ensuring the uniformity of gas/liquid flow distribution. Typically, about 5 g of catalyst extrudates were loaded in the reactor as per the procedure. The catalyst was dried under nitrogen flow at 100 mL/min for 12 h at a temperature of 120 °C followed by reduction for a period of 5 h at 300 °C. The system pressure and temperature were brought to the desired operating conditions, and the HDO feed was pumped at the rate of 7.2 mL/h in the reactor using a high-performance liquid chromatography pump (HPLC pump). The product from the reactor outlet was cooled using a heat exchanger connected to a chiller, and it was further separated into gas and liquid phases in a separator. Liquid products were collected and subjected to simulated distillation (SIMDIST) analysis. Further D-86 analysis was carried out on the liquid products for obtaining the hydroisomerised Kero fraction (150–300 °C). This pour-point of this fraction was measured following the ASTM D97 method.

3. Results and discussion

3.1. Catalyst characterisation

3.1.1. X-ray diffraction

The acquired X-ray diffraction (XRD) patterns covered an angular range of 2θ = 10–80°. The patterns were recorded with a step size of 0.01° and a step acquisition time of 0.5 s. The average crystallite size, L, was calculated using the Scherrer Eq. (1).

$$L = K\lambda/\beta cos(\theta) \quad (1)$$

where K is a dimensionless shape factor, with a value close to unity; λ is the wavelength of the X-ray radiation; β is the line broadening at half the maximum intensity (FWHM), and θ is the Bragg angle. Instrument broadening was considered.

X-ray diffraction (XRD) analysis provided insights into the crystallinity and distribution of NiMo catalysts supported on various materials. Despite **γ**-Al_2O_3 predominant amorphous nature and well-dispersed active species, subtle distinctions became apparent with increasing silica content. The catalyst with Al_2O_3/SiO_2 ratio of 90:10 showed faint signals suggesting residual silica, while the catalyst with Al_2O_3/SiO_2 ratio of 70:30 catalyst exhibited a broad hump indicative of amorphous silica. Crystalline phases of MoO_3, NiO, and their interaction product ($NiMoO_4$) remained elusive, particularly in the high-silica catalyst. The crystallite size as estimated for Ni from the Scherrer Eq. (1) is 4 nm possibly existing as nanoscale crystallites (sub-4 nm) or well-dispersed species on the high surface area supports. No $NiMoO_4$ peaks suggested a potential low abundance or potential amorphous state.

The peaks observed around 2θ values of approximately 37.5°, 45.9°, and 67.5°; corresponding to the (311), (400), and (440) crystallographic planes, respectively; confirm the gamma alumina. The peak around 21° corresponds to the SiO_2 (101) inclusion in 90:10 and 70:30 Alumina: Silica supports. The incorporation of more silica has increased the crystallinity of the support as reflected in 70:30 Alumina: Silica support. There are no evident peaks of Ni and Mo in the XRD pattern of the catalyst indicating higher dispersion and a high amorphous nature of the support.

To further investigate the catalyst's composition and validate the good dispersion of metal oxides on the support we performed XRD analysis of the catalyst at higher temperatures (up to 1000°C). Fig. 3. Well-dispersed metal oxides like NiMo on Al_2O_3-SiO_2 (70:30) support typically exhibit good stability at calcination temperatures up to 750 °C.

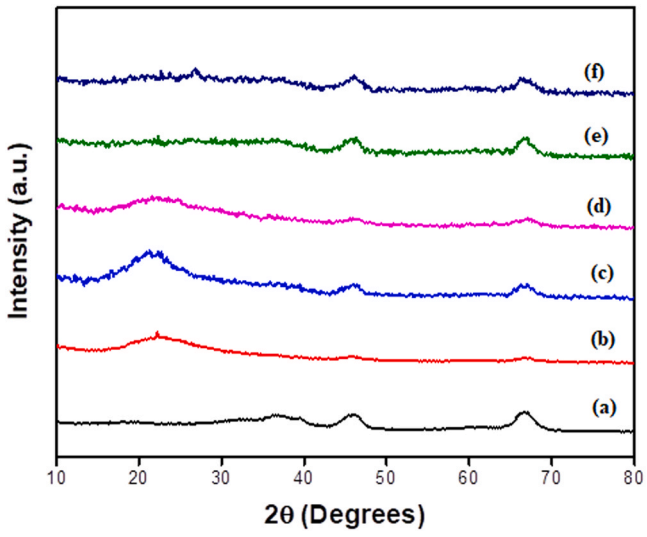


Fig. 1. X-ray analysis of (a) γ-Al_2O_3 (b) Al_2O_3-SiO_2 (90:10) (c) Al_2O_3-SiO_2 (70:30) (d) NiMo/γ-Al_2O_3 (e) NiMo/ Al_2O_3-SiO_2 (90:10) (f) NiMo/ Al_2O_3-SiO_2 (70:30).

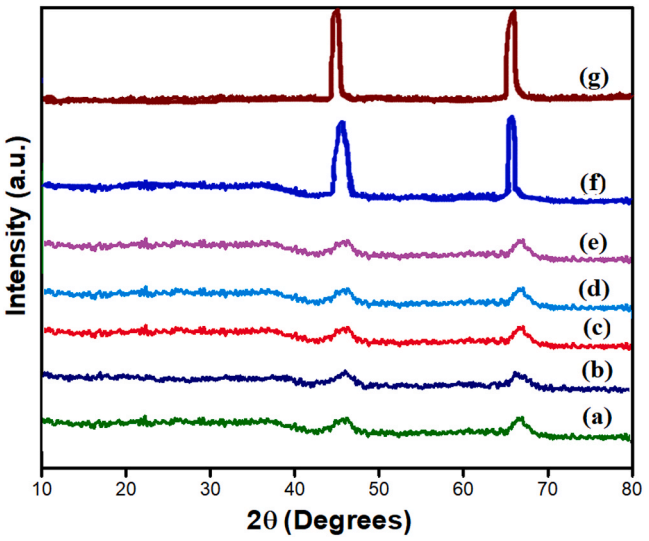


Fig. 2. X-ray analysis of NiMo/Al_2O_3-SiO_2 (70:30) at (a) 550 °C (b) 600 °C, (d) 650 °C (e) 700 °C (f) 750 °C (g) 900 °C (h) 1000 °C.

Interestingly, we observed an increase in the intensity of the peaks around 46° and 66.7° at higher temperatures (900 & 1000 °C). This could suggest the presence of (111) and (211) planes of Ni and Mo species, respectively. Additionally, Scherrer equation analysis of the XRD pattern revealed a crystallite size of approximately 60 nm for the intense metal oxide peaks, further supporting the presence of highly dispersed nanoparticles at temperatures below 550 °C. High temperatures can induce phase transformations in the metal oxides, influencing their crystallinity properties. Optimizing the calcination process requires balancing these factors to achieve well-dispersed and catalytically active nanoparticles.

3.1.2. *Textural properties*

Nitrogen adsorption-desorption isotherms (Fig. 4) provide valuable insights into the textural properties of γ-Al_2O_3 and Al_2O_3-SiO_2 supports. All samples exhibit type IV isotherms, indicating mesoporous structures with narrow pore size distributions. Hysteresis loops confirm the presence of mesopores, particularly for γ-Al_2O_3, Al_2O_3-SiO_2 (90:10) and Al_2O_3-SiO_2 (70:30). Specific surface area was calculated from adsorption data in the range ($0.01 < P/Po < 0.08$). The surface area decreases with increasing SiO_2 content: γ-Al_2O_3 (250.14 m^2/g) > Al_2O_3-SiO_2 (90:10) (224.30 m^2/g) > Al_2O_3-SiO_2 (70:30) (190.52 m^2/g) given in Table 1. The observed decrease in surface area with increasing SiO_2 content in Al_2O_3-SiO_2 supports can be attributed to the modification of

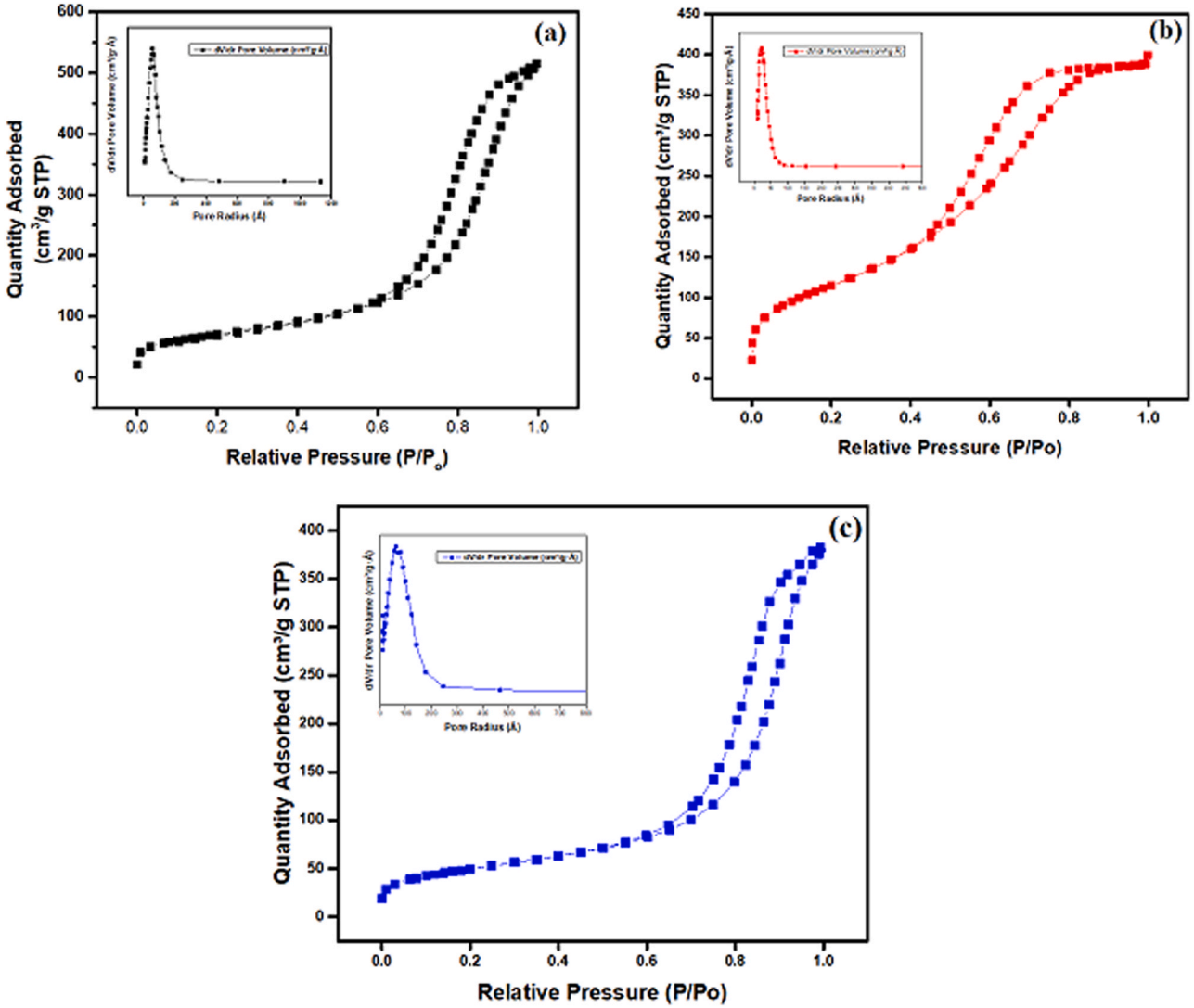


Fig. 3. BET analysis of (a) γ-Al_2O_3 (b)Al_2O_3-SiO_2 (90:10) (c)Al_2O_3-SiO_2 (70:30).

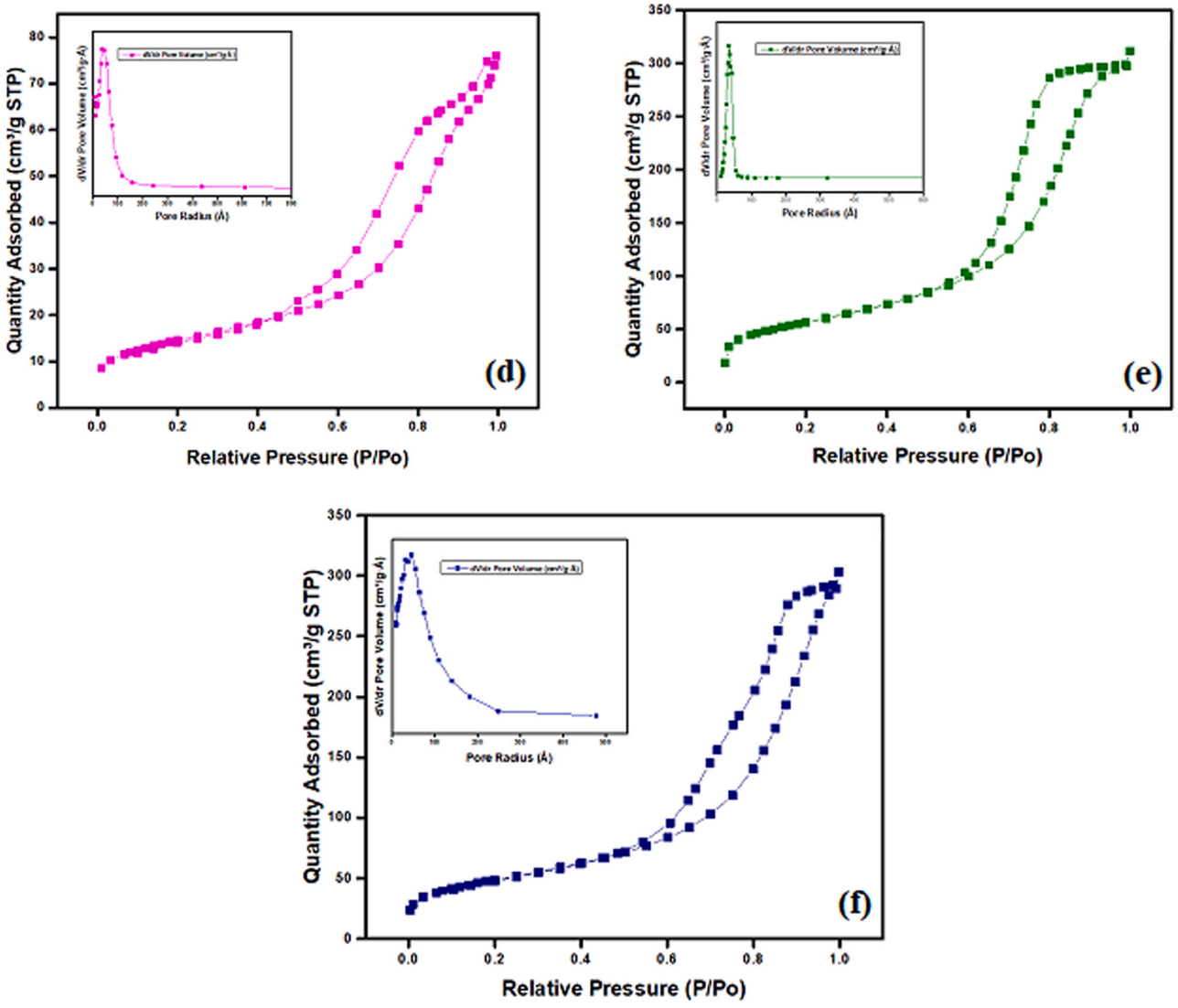


Fig. 4. BET analysis of (d) NiMo/γ-Al_2O_3, (e) NiMo/Al_2O_3-SiO_2 (90:10), (f) NiMo/Al_2O_3-SiO_2 (70:30).

Table 1
Surface Area and pore size distribution analysis.

Catalyst	S_{BET} m^2/g	BJH Total Pore Volume cm^3/g	BJH Average Pore Diameter Å
γ-Al_2O_3	250.1	0.60	52.85
Al_2O_3: SiO_2 (90:10)	224.3	0.65	62.16
Al_2O_3: SiO_2 (70:30)	190.5	0.80	68.85
NiMo/γ-Al_2O_3	210.3	0.50	45.28
NiMo/Al_2O_3: SiO_2 (90:10)	171.5	0.49	45.55
NiMo/Al_2O_3: SiO_2 (70:30)	151.79	0.45	41.28

the alumina's pore structure by silica. This phenomenon likely occurs through a combination of factors. Firstly, SiO_2 particles may act as pore fillers, occupying the voids within the Al_2O_3 matrix and reducing the accessible surface area. Secondly, silica deposition can lead to a denser and smoother surface, diminishing the inherent roughness of alumina and thereby decreasing the overall surface area available for adsorption. Additionally, the presence of SiO_2 might influence the pore size distribution, potentially hindering the formation of smaller mesopores which contribute more significantly to the total surface area. Finally, during synthesis, SiO_2 could promote the aggregation of alumina particles, resulting in larger clusters with a lower total surface area compared to the sum of individual particles. These combined effects of pore filling, surface smoothing, altered pore size distribution, and particle aggregation contribute to the observed decrease in surface area with increasing SiO_2 content in Al_2O_3-SiO_2 supports [29]. The total pore volume (assessed at P/Po = 0.95) and average pore diameter (BJH method) increase with increasing SiO_2 content. γ-Al_2O_3 shows the smallest pore volume (0.60 cm^3/g) and narrowest average pore diameter (52.85 Å), indicating a more confined network. Higher SiO_2 content leads to pore structures with larger pore volumes and diameters. The surface areas and pore volumes of the impregnated active catalysts varied in the range (151–210 m^2/g and 0.4–0.5 mL/g, respectively), highlighting the influence of metal deposition on textural properties.

3.1.3. *Ammonia temperature programmed desorption (NH_3-TPD)*

The ammonia temperature-programmed desorption (NH_3-TPD) results highlight the critical role of the Al_2O_3:SiO_2 ratio in fine-tuning the acidity of catalyst supports and NiMo bimetal catalysts, which significantly impacts the selectivity and conversion of sustainable aviation fuel (SAF) range hydrocarbon molecules. The γ-Al_2O_3 support exhibits a notable presence of weak and strong acid sites, while the introduction of SiO_2 in the Al_2O_3:SiO_2 mixtures enhances both weak and strong acid

sites. The Al_2O_3:SiO_2 (70:30) mixture, in particular, displays a substantial increase in weak and strong acid sites, making it a promising candidate for catalyzing SAF range hydrocarbon molecules.

As seen from Table 2, in the context of the supports, γ-Al_2O_3 demonstrates a substantial presence of weak and strong acid sites with 98.9 μmol/g and 12.7 μmol/g, respectively. Interestingly, the incorporation of SiO_2 into the supports, as seen in the NiMo/ Al_2O_3:SiO_2 (90:10) and (70:30) catalysts, reveals distinct trends. The 90:10 mixture shows an augmentation in weak acid sites (305 μmol/g) with minimal impact on strong acid sites (76.1 μmol/g), while the 70:30 blend exhibits heightened weak acid sites (645.4 μmol/g) alongside a reduction in strong acid sites (85.6 μmol/g). Turning to the NiMo bimetal catalysts, NiMo/ γ-Al_2O_3 reveals a minimal increase in weak acid sites (101.2 μmol/g) but significant amounts of strong acid sites (81.8 μmol/g).

Gamma alumina (γ-Al_2O_3) is a widely used catalyst material known for its acidic properties. The acidity of SiO_2-Al_2O_3 catalysts arises from the presence of both Brønsted and Lewis acid sites [30,32]. Brønsted acid sites are typically associated with the presence of hydroxyl groups (–OH) on the surface, which can donate protons (H^+). These sites are often formed through the interaction of SiO_2 with Al_2O_3, creating Si–OH–Al linkages that exhibit Brønsted acidity [31]. On the other hand, Lewis acid sites are associated with coordinatively unsaturated aluminum atoms that can accept electron pairs [31–33]. Silica (SiO_2) in the γ-Al_2O_3 matrix significantly impacts the overall acidity of the resulting composite material [31].

Several factors contribute to the increased acidity observed in silica-alumina (SiO_2-Al_2O_3) compared to pure alumina (Al_2O_3). This enhancement can be attributed to the synergistic interaction between silica and alumina, as demonstrated by Baiker and Huang groups [34]. Bokhoven et al. [35] explored the correlation between Lewis acid activity and the presence of extra-framework aluminum species in zeolite Y. Their work suggests that incorporating silica into the alumina framework increases the dispersion of aluminum atoms. This dispersion creates more exposed aluminum sites, which act as Lewis acid centers. Furthermore, the formation of Si–OH–Al bridges introduces new Brønsted acid sites [36]. Notestein et al. [37] observed the unexpected formation of these Brønsted acid sites upon adding SiO_2 to alumina. Additionally, silica acts as a structural promoter, altering the distribution and strength of acid sites on the Al_2O_3 support [35–37]. These mechanisms are well documented in the literature.

The observed trends in acidity for γ-Al_2O_3 and its composites with varying SiO_2 content (Table 2) show a fascinating trend of how the material's composition influences its acidic properties. As expected from its inherent structure, pure γ-Al_2O_3 exhibits the presence of both weak and strong acid sites. This reflects the coexistence of Lewis and Brønsted acidity on its surface. When a small amount of SiO_2 (10 %) is introduced, there's a noteworthy increase in weak acid sites observed in γ-Al_2O_3:SiO_2 (90:10) catalyst composition. This phenomenon can likely be attributed to the formation of new Brønsted acid sites at the interface between the Al_2O_3 and SiO_2 phases, as discussed earlier. Interestingly, the strong acid sites remain relatively unchanged. This suggests that the 10 % SiO_2 content might cause a slight decrease in the concentration of Al^{3+} cations (which are responsible for Lewis acidity), but it's not enough to significantly affect the inherent strong acid character of γ-Al_2O_3.

When the SiO_2 content is increased to 30 % (γ-Al_2O_3:SiO_2 (70:30)) a sharp increase in weak acid sites is observed, likely due to a more prominent formation of new Brønsted acid sites at the interface compared to the 90:10 ratio. However, the higher concentration of SiO_2 leads to a more pronounced decrease in the number of accessible Al^{3+} sites. This decrease in accessible sites, coupled with the potential modification of existing sites by SiO_2, results in a lower overall acidity compared to pure γ-Al_2O_3 and the 90:10 mixture [31].

The incorporation of NiMo into these catalysts further influences their overall acidity. While NiMo itself might not possess significant inherent acidity, its interaction with the acidic support plays a crucial role. The observed changes in acidity upon NiMo incorporation likely depend on the interplay between the acidic support (whether pure γ-Al_2O_3 or Al_2O_3-SiO_2 composite) and the deposited NiMo species. This interplay can affect the distribution and strength of acid sites on the catalyst surface, potentially leading to further modifications in overall acidity.

Table 2
Acidity measurement NH_3-TPD analysis.

Catalyst	Weak Acid Sites (<300°C) (μmol/g)	Strong Acid Sites (>500°C) (μmol/g)	Total Acidity (μmol/g)
γ-Al_2O_3	98.9	12.7	111.6
Al_2O_3:SiO_2 (90:10)	244.4	67.6	312.0
Al_2O_3:SiO_2 (70:30)	358.9	276.1	635
NiMo/γ-Al_2O_3	101.2	81.8	183.0
NiMo/Al_2O_3: SiO_2 (90:10)	305	76.1	381.1
NiMo/Al_2O_3: SiO_2 (70:30)	645.4	85.6	740.0

3.2. Evaluation of catalytic activity

Hydrodeoxygenation of UCO

3.2.1. Comparison of catalytic activity with different supports

The reaction products obtained were analysed for their physical properties. The viscosity of the reaction products obtained was in the

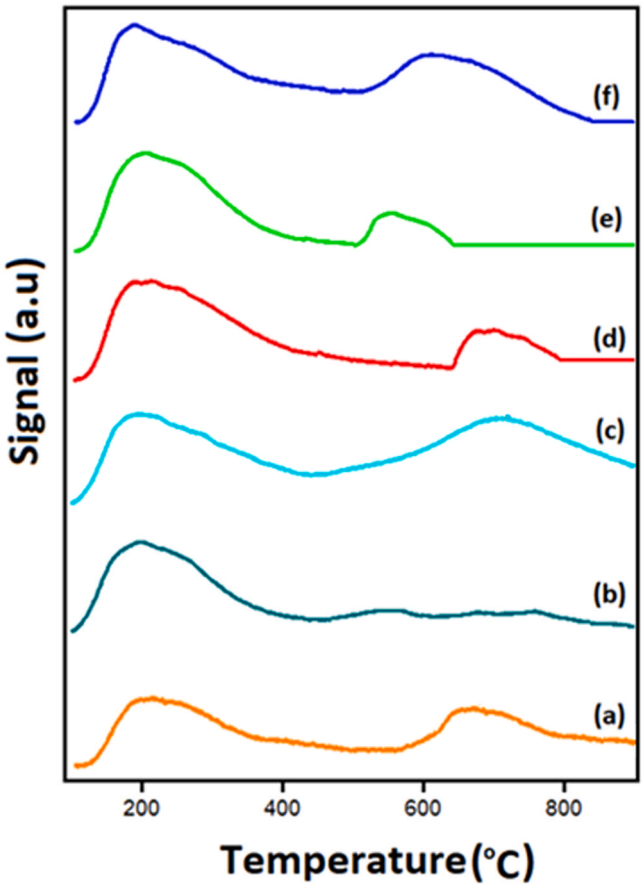


Fig. 5. NH_3-TPD analysis of (a) γ-Al_2O_3 (b) Al_2O_3-SiO_2 (90:10) (c) Al_2O_3-SiO_2 (70:30), (d) NiMo/γ-Al_2O_3, (e) NiMo/Al_2O_3-SiO_2 (90:10), (f) NiMo/Al_2O_3-SiO_2 (70:30).

range of 3.00–3.20 Cst as compared to a viscosity of 43 Cst for the feed UCO. There is a decrease in the FFA content and bromine no of the product stream. This decrease in the viscosity, FFA content, and bromine no of the product stream is a clear indication of HDO. The reaction product obtained was also subjected to SIMDIST analysis to check for the fractions of gasoline, kerosene, diesel, and unconverted UCO. The unconverted UCO was measured to obtain the conversion of UCO during the HDO reaction. The conversion of UCO increases with an increase in the acidity of the support [23,24]. NiMo/ γ-Al_2O_3 (CAT A) prepared using alumina as the support is less active and therefore shows less conversion of UCO. The conversion of UCO increases from 80 % on CAT A to 85 and 87 % on catalysts NiMo/Al_2O_3-SiO_2 (90:10) (CAT B) & NiMo/Al_2O_3-SiO_2 (70:30) (CAT C) respectively prepared with alumina-silica as the support. This is because alumina-silica supports are more acidic than pure gamma-alumina supports. The distribution of various fractions obtained from SIMDIST analysis is also shown in Table 3. The CAT-A is more diesel selective while catalysts B&C are more selective to kerosene. There is a substantial increase in kerosene fraction from 45 % on CAT-A to 75 % on CAT-C with 30 % silica-incorporated alumina support. This again proves that the increase in the acidity of the catalysts results in higher cracking activity leading to increased kerosene selectivity. There is not much increase in the gasoline content of the product stream with acidity, thus proving that the catalysts B&C have intermediate acidity to increase the cracking activity for only obtaining higher kero selectivity.

The structure-activity relationship (SAR) in catalysis is a crucial aspect of understanding the performance of a catalyst in a given reaction. It refers to the correlation between the structural features of a catalyst and its activity, selectivity, and stability.

In the context of the study on highly selective hydrodeoxygenation for sustainable aviation fuel production from used cooking oil, the SAR plays a significant role in optimizing the catalyst activity and selectivity for converting used cooking oil (UCO) into jet-range hydrocarbons. Our study demonstrated the structures of pure alumina, different ratios of alumina-to-silica in the catalyst support play a crucial role in influencing the interactions and activity of converting UCO into jet-range hydrocarbons.

i. Pure Alumina Structure: Pure alumina exhibits specific textural properties, such as a surface area of 250.14 m^2/g and a pore volume of 0.60 cm^3/g with a narrow average pore diameter of 52.85 Å. This structure indicates a more confined network with smaller pores, influencing the catalyst's surface area and pore characteristics.
ii. Different Ratios of Alumina-Silica: Varying the alumina-to-silica ratios in the catalyst support impacts the textural properties. For instance, the Al_2O_3-SiO_2 (90:10) support shows a surface area of 224.30 m^2/g and a larger pore volume compared to pure alumina. As the silica content increases, the pore volume and average pore diameter also increase, leading to structures with larger pores and volumes.
iii. Alumina-to-Silica Ratios in the Catalyst Support: The study identifies a 70:30 alumina-to-silica ratio as optimal for achieving exceptional UCO conversion (86.69 %) and jet range hydrocarbon fuel selectivity (74.79 %). This specific ratio influences the catalyst's activity and selectivity, highlighting the importance of the silica-to-alumina ratio in determining the performance of the catalyst in converting UCO into jet-range hydrocarbons.

The structures of pure alumina and different ratios of alumina-silica in the catalyst support significantly impact the textural properties, surface area, pore volume, and pore diameter of the catalysts. These structural variations play a crucial role in determining the interactions and activity of the catalyst in converting UCO into jet-range hydrocarbons, ultimately influencing the efficiency and selectivity of the hydrodeoxygenation process for sustainable aviation fuel production.

The UCO conversion and selectivity were calculated based on the formula given below.

$$X_{Feed}(\%) = \left[\frac{F_I - F_O}{F_I}\right] * 100$$

$$S_{Kero}(\%) = \left[\frac{K_O}{F_I - F_O}\right] * 100$$

$$S_I(\%) = \left[\frac{I_O}{F_I - F_O}\right] * 100$$

Where X_{Feed} is feed conversion, F_I is feed inlet flow rate (g/h), FO is feed outlet flow rate (g/h), K_O is kerosene outlet flow rate (g/h) and I_O is outlet flow rate (g/h) of any product species 'I' forming in the reaction and 370 + ^{0}C boiling fraction in hydro-deoxygenated oil is considered to be unreacted feed.

Table 3
SIMDIST analysis of product obtained from hydrodeoxygenation of used cooking oil.

Catalysts	Active components	Al_2O_3	SiO_2	Viscosity @ 40 C (Cst)	PP°C	Br No.	Product selectivity (%)						UCO Conv. (%)
							FFA	Gas	Gasoline (36–150 C)	Kero (150–300 C)	Diesel (300–370 C)	Water	
CAT A	NiMo: (P: 0.5 %, Ni: 3 %, Mo: 10 %)	100	0	3.08	15	1.27	0.28	1.35	4.77	45.00	40.28	7.16	80.24
CAT B	NiMo: (P: 0.5 %, Ni: 3 %, Mo: 10 %)	90	10	3.11	12	0.79	0.002	6.02	1.28	69.20	6.84	7.32	85.42
CAT C	NiMo: (P: 0.5 %, Ni: 3 %, Mo: 10 %)	70	30	3	15	0.64	0.016	5.26	1.41	74.79	10.75	8.02	86.69

PP: Pour point; Br. No.: Bromine Number; FFA: Free Fatty Acid and UCO: Used Cooking Oil
Reaction conditions: Temperature, 350°C, Flow rate, 1.1 h^{-1} WHSV, Pressure, 6 MPa, H_2/oil ratio, 400

3.2.2. Comparison of catalytic activity with different catalysts

Table 4 presents a summary of various catalysts and their corresponding products, including hydrocarbon fuel and green diesel. In Table 5, the NiMo catalyst, which is supported on Alumina-Silica with an Al_2O_3/SiO_2 ratio of 70:30, demonstrates a remarkable UCO conversion rate of 86.69 % during the hydrodeoxygenation process. Additionally, this catalyst exhibits a significant 74.79 % selectivity towards jet-range hydrocarbon fuels under mild conditions (350°C, 1.1 h^{-1} WHSV, 6 MPa, and H_2/oil 400). These results highlight the potential of the Al_2O_3-SiO_2 catalyst for converting UCO into SAF-range hydrocarbons. These findings underscore the significant potential of the NiMo/ Al_2O_3-SiO_2 catalyst for transforming UCO into SAF range hydrocarbons but also emphasize the importance of selecting appropriate catalysts and process parameters, such as WHSV (Weight Hourly Space Velocity), to optimize conversion efficiency and target the desired C_9-C_{15} product range for sustainable aviation fuels.

Origin of Catalytic Active Sites: The catalytic activity of NiMo-supported catalysts on Al_2O_3 and SiO_2-Al_2O_3 supports is primarily due to the presence of both metal active sites and acidic sites.

(i) Metal active sites: The Ni and Mo atoms on the catalyst surface form active sites essential for hydrogenation and hydrodeoxygenation reactions [47]. These sites are created during the impregnation and calcination processes. These metal active sites facilitate the activation of hydrogen and its subsequent reaction with unsaturated hydrocarbons, leading to the production of hydrocarbon biofuels.
(ii) Acidic sites: Brønsted acid sites formed by Si–OH–Al linkages in SiO_2-Al_2O_3 supports [48], providing protons necessary for cracking and isomerization reactions. Lewis acid sites associated with coordinatively unsaturated aluminum atoms, these sites are crucial for the adsorption and activation of reactant molecules.
(iii) Support effect: The incorporation of SiO_2 into Al_2O_3 modifies the support's acidic properties and enhances the dispersion of NiMo particles [49,50]. This leads to a higher concentration of active sites and stronger acid sites in SiO_2-Al_2O_3 supports. The better dispersion of NiMo particles on SiO_2-Al_2O_3 supports increases the number of accessible active sites, contributing to the observed catalytic performance.

Despite the marginal differences in catalytic results, the slight performance enhancement observed for NiMo/SiO_2-Al_2O_3 catalysts can be attributed to improved dispersion of NiMo particles, increased number and strength of acidic sites, enhanced metal-support interactions. These factors collectively contribute to the overall catalytic performance, explaining the slight differences observed in Tables 4 and 5.

3.2.3. Catalyst Stability studies

All three catalysts were subjected to long-term testing to check the stability of the catalysts. The data is plotted in Fig. 6. The figure shows that all three catalysts show a stable trend of activity to UCO conversion as well as kerosene selectivity for up to 120 hours of the run without the requirement of catalyst re-sulfidation. This shows that the catalysts B&C prepared with silica-alumina supports are stable for hydrodeoxygenation to produce kerosene selective fraction during the HDO of UCO. All the catalysts were prepared by loading phosphorus on the support followed by impregnation of molybdenum and nickel. Phosphorus is considered the second promoter of hydrotreating catalysts. The phosphorus increases the catalytic activity and reduces the deactivation of catalysts by preventing the formation of inactive $NiAl_2O_4$ particles. It also improves the thermal stability of alumina and prevents coke formation during hydrotreating reaction thus ensuring enhanced catalyst life [25–27].

3.3. Hydroisomerisation of Hydrodeoxygenated Oil (HDO)

Typically, hydroisomerization catalysts are bifunctional in nature and use of a noble metal such as platinum (Pt) for the dehydrogenation/hydrogenation function over acidic support such as medium pore zeolites like ZSM-22, ZSM-23, and ZSM-48 are reported in the literature. These catalysts with dual functionality are used to perform isomerization reaction of the olefinic intermediates formed over the metal sites followed by branching over the acidic sites of the zeolite support during hydroisomerization of long chain n-paraffin. The dried HDO obtained from the optimized catalyst formulation CAT C (NiMo/ Al_2O_3-SiO_2 (70:30) was subjected to hydroisomerization and mild cracking for the production of jet fuel. The results of the production of Jet-Fuel from Hydrodeoxygenated Oil (HDO) are shown in Table 6. The conversion was carried over one-dimensional zeolite Pt/H-ZSM-22 based on our previous work on hydroisomerization of n-hexadecane at a temperature range of 310–330 °C based on our previous work on hydroisomerization of n-hexadecane [28]. A pour point of −60 °C was obtained for the product after the hydroisomerization and mild cracking of HDO at a temperature of 325 °C. Therefore, the (NiMo/ Al_2O_3-SiO_2 (70:30)) catalyst produces a feed with the required specification to obtain jet range product in the isomerization reaction.

4. Summary and conclusion

NiMO catalyst supported on Al_2O_3-SiO_2 mixed metal oxide showed superior performance as compared to Al_2O_3 only supported catalyst. High surface area, mesoporous Al_2O_3-SiO_2 mixed oxide supports were synthesized using a sol-gel homogeneous co-precipitation method,

Table 4
Catalytic conversion of various vegetable oils into Biofuels.

Vegetable Oil	Catalyst	Conditions	Reactor	Main Product	Product Yield (wt %)	Ref
Sunflower	Co-Mo/ Al_2O_3	300–380℃, 2–7.9 MPa	High pressure reactor	Organic product	83.4–93.3	[38]
Refined sunflower	Ni-Mo/ Al_2O_3	360℃, 3 MPa	High pressure-trickle bed reactor	Green diesel	-	[39]
Canola	Ni-Mo/SiO_2-Al_2O_3	350℃, 3.9 MPa	High pressure fixed- bed reactor	Bio-hydrogenated diesel	81.4	[40]
Carinata	Mo-Zn/ Al_2O_3	350℃, 2 MPA	Batch reactor	Hydrocarbon biofuel	92.6–93.5	[41]
Soybean	NiMo/ZSM-5 carbide; NiMo/ZSM-5 nitride	360–450℃, 4.4 MPa	Fixed-bed reactor	Hydrocarbon fuel	50	[42]
Karanja	Ni / Al_2O_3	380℃, 3.5 MPa	Semi-batch reactor	Green diesel	80	[43]
Used Cooking Oil (UCO)	NiMo/ Al_2O_3	350℃, 60 MPa	Fixed bed reactor	Hydrocarbon biofuel	92	Current Work
Used Cooking Oil (UCO)	NiMo/SiO_2- Al_2O_3 (10/90)	350℃, 60 MPa	Fixed bed reactor	Hydrocarbon biofuel	93.6	Current Work
Used Cooking Oil (UCO)	NiMo/SiO_2 – Al_2O_3 (30/70)	350℃, 6 MPa	Fixed bed reactor	Hydrocarbon biofuel	93.7	Current Work

Table 5
Comparison of process parameters for converting various vegetable oils into sustainable aviation fuel hydrocarbons.

Feedstock	Catalyst	Weight Hourly Space Velocity (h^{-1})	Reactor	Mass Balance (wt%)	Conversion Rate (wt %)	Ref
Carinata oil	Mo-Zn/Al_2O_3	-	Fixed-bed reactor	95–96	-	[41]
Waste vegetable oil	NiMo/Al_2O_3, CoMo/Al_2O_3	2.8	Autoclave reactor	-	82	[44]
Vegetable oil	NiMo/ Al_2O_3, Mo/ Al_2O_3, NiMo/ Al_2O_3	0.25–4	Fixed-bed reactor	-	30–100	[45]
Vegetable oil	CoMo/Ordered Mesoporous Alumina (OMA)	1.5–2	Fixed-bed reactor	-	75–95	[46]
Used Cooking Oil (UCO)	NiMo/ Al_2O_3	1.2	Fixed-bed reactor	>98	80.24	Current Work
Used Cooking Oil (UCO)	NiMo/SiO_2- Al_2O_3 (10/90)	1.2	Fixed-bed reactor	>98	85.42	Current Work
Used Cooking Oil (UCO)	NiMo/SiO_2 – Al_2O_3 (30/70)	1.2	Fixed-bed reactor	>98	86.69	Current Work

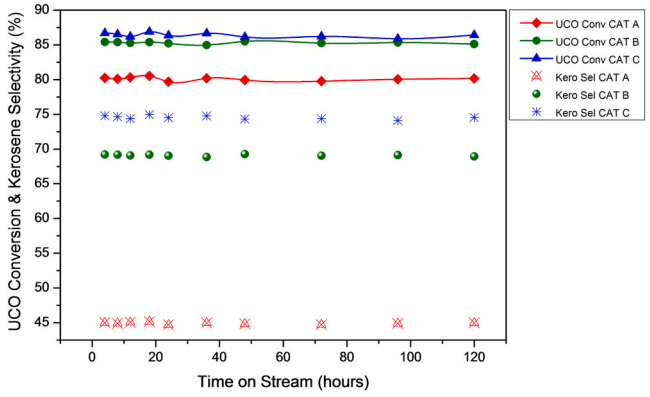


Fig. 6. Catalyst stability studies of Cat A: NiMo/γ-Al_2O_3, CAT B: NiMo/Al_2O_3-SiO_2 (90:10), and CAT C: NiMo/Al_2O_3-SiO_2 (70:30).

Table 6
Hydroisomerization and mild cracking of HDO to Jet fuel.

	SIMDIST profile				Pour Point
Temp (°C)	<=150 (°C)	150–300 (°C)	260–320 (°C)	320+ (°C)	150–300 (°C)
315	11	86	50.5	0	-48
325	18.5	79	41	1	-60

ensuring a nanoscale distribution of Al and Si elements. The incorporation of SiO_2 into the Al_2O_3 support modifies the textural properties and metal-support interactions, consequently impacting the catalytic activity and formation of catalytic sites. The introduction of SiO_2 significantly altered these properties and their interaction with MoS_2 phases, ultimately impacting the catalyst activity for hydrodeoxygenation of used cooking oil (UCO).

The experimental results also revealed that tuning of the Al_2O_3-SiO_2 ratio is a crucial factor in optimizing the textural properties of support, metal-support interactions, and the formation of catalytic sites. This tailored design enhances the catalytic performance to produce sustainable aviation fuel range hydrocarbons. Notably, an optimal increase in the SiO_2 content promoted the selective hydrodeoxygenation of UCO, yielding a higher fraction of kerosene-range hydrocarbons suitable for further refinement into jet fuel with a pour point of −60°C. In conclusion, the potential of Al_2O_3-SiO_2 mixed oxide supports for the selective conversion of pretreated UCO feedstocks into sustainable aviation fuel components has been experimentally validated through high catalyst activity and long-term stability of the catalyst.

CRediT authorship contribution statement

Lalit Kumar: Formal analysis, Data curation. **Shivanand Pai:** Writing-review & editing, Formal analysis, Data curation. **Arundhathi Racha:** Writing-original draft, Writing-review & editing, Visualization, Conceptualization, Data curation, Formal analysis, Validation, Resources, Methodology, Investigation, Software, Resources. **Chiranjeevi Thota:** Supervision, Project administration. **Chanchal Samanta:** Writing – review & editing. **Bharat L Newalkar:** Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Acknowledgments

The Authors thank BPCL management for approval to publish this work.

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